



Fostering innovation under institutional deficiencies: formal state–business consultation or cronyism?

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Abstract

This paper is trying to investigate whether formal state–business consultations (or simply public private dialogues—PPD) and/or cronyism could counteract low institutional capacities and provide the needed mechanisms for fostering innovation. It is doing this with the help of a theoretical discussion highlighting the links between National Innovation System, state business relations (SBR) and good governance institutions. This is then followed by an empirical study based on this discussion and using various OLS multivariate panel regressions. Evidence is found that the presence of business associations that are widely participated help in promoting innovation. They do this also in countries suffering from low institutional settings, and correct for the shortcomings caused by low regulatory quality. PPD play also an important role in counteracting institutional deficiencies that could otherwise harm innovation; and this is specifically the case for low government effectiveness. High official representation in PPD is a key for such needed outcome. PPD, however, need an open climate where the government is accountable to various societal groups and where different interest groups are free to aggregate their power. On the other hand, favoritism in allocation of resources and Cronyism might counteract institutional deficiencies, especially with regard to low government effectiveness and low regulatory quality. Yet, the overall effect of cronyism on innovation is still negative. Tackling the reduction of cronyism as an objective presents PPD as an alternative that could counteract low government effectiveness and high participation in business associations as another alternative which manages to counteract low regulatory quality, provided that cronyism is minimized in both cases. The novelty of this study rests on the contribution it aims at providing in an area where research is insufficient. There is lack of research on how innovation is affected generally by SBR, and more specifically by either formal or informal forms of SBR and cronyism. Empirical studies are even scarcer in this regard.

Keywords Innovation, national-innovation-system · State–business-relations · Public–private-dialogue · Cronyism · Institutions

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1 Introduction

It is nowadays largely beyond doubt that innovation and technological upgrading are important determinants of economic growth, an argument supported by a big body of literature and empirical evidence. A prominent part of this literature is presented in endogenous growth theories such as “Schumpeterian” models, which suggest that an economy’s growth rate is increasing with an increase in Research and Development (R&D) productivity and is positively correlated with the increase in patents, rate of firms’ entry and exit and the introduction of new products (Aghion and Howitt 1998, p. 434). This is further supported by various empirical studies showing a positive relation between innovation and economic growth (see for instance: Capello and Lenzi 2014; Ulke 2004). Even when acquiring foreign technology, domestic innovational efforts (e.g.: R&D spending) are still regarded as essential for the sake of creating “absorptive capacity” allowing better use of available technology (Cohen and Levinthal 1989, 1990). Lack of in-house innovational endeavors hinders learning by doing in many developing countries (El-Elj 2012, p. 187). Romijn and Caniëls (2011) suggested that developing “domestic capabilities for creative innovation” which addresses local needs is required, especially now in a world faced by “unprecedented challenges” (e.g.: resource depletion, population pressure, etc.).

But, if innovation plays a decisive role for economic growth, how could it be fostered? Intuition tells us that good policies and regulations are needed to realize this objective. These could range from providing tax incentives, to increasing public investment in education and training, to easing credit allocation for R&D, to many other possible alternatives. Yet, efficiently formulating and conducting such innovation-related policies and regulations necessitate on their turn the presence of good institutions. In terms of Good Governance terminologies, this would translate into high regulatory quality and government effectiveness. Nevertheless, such needed institutional capacities are often missing in different societies; and institutional development needs substantial resources and time (Enderwick 2005, pp. 119–120). This poses the need for finding another mechanism that could substitute and counteract such institutional deficiencies, a mechanism that would efficiently formulate and implement policies and regulations needed for fostering innovation. It is based on these concerns that intensive consultations stemming from closer State Business Relations (SBR) could be suggested to provide such a mechanism. It is suggested in the literature on SBR that they could be helpful in addressing business and government failures, building trust, fostering exchange of information between the government and businesspeople as well as helping in the formulation of better policies (see Sabry 2013, 2017b; Leftwich and te Velde 2010; Hausmann et al. 2008; Haggard et al. 1997 among others). All of these are relevant for efforts needed to foster innovation. In developing countries, often suffering from various institutional deficiencies, the need for state–business consultation seems to be even higher.

There is the risk, however, that close SBR could encourage cronyism (see Sabry 2013, 2017a, b), where the benefit in terms of innovation and more generally

economic improvement could be minimal and/or confined to few connected businesspersons. Cronyism is preferential treatment provided by officials to businesspersons (or others) who are socially (e.g.: old friends and associates) or politically connected to them, regardless to businesspersons' qualifications (Begley et al. 2010; Mazumdar 2008). Cronyism is widely regarded as having negative economic consequences, for instance, because of distorting economic incentives and augmenting transaction costs. Yet, some scholars argue that cronyism, on the contrary, lower transaction costs, help in the circulation of better information, agreement enforcement, and cost reduction, and eventually lead to economic growth (Kang 2003). If such claims are true, cronyism could provide a mechanism for fostering innovation, even if it benefited only the few connected. Again, arguably, such a mechanism could be needed more when institutional deficiencies exist.

This presents us with two alternatives for fostering innovation especially in societies suffering from institutional deficiencies. The first is formal state–business consultations. These are councils where representatives from the government and businesspeople (and sometimes others) convene, discuss issues of concern and formulate policies. Formal state–business consultation could refer to Social and Economic Councils, business forums and Public Private Dialogues (PPD).¹ The second alternative is cronyism, which is more likely induced by informal SBR (Sabry 2013, 2017b), or relations based on social (familial, tribal, etc.) and political (e.g.: common party membership) connections between state officials and businesspersons. When privileges—such as subsidies, credit allocations, and market domination—are provided for those socially and politically connected businesspersons, cronyism is the ultimate result. Arguably, if cronyism could truly facilitate information exchange, reduce transaction costs and act as a mechanism for consultation (even if it is benefiting the few), it could possibly then counteract institutional deficiencies that hinder formulating policies and regulations fostering innovation.

Based on these considerations this paper is presented. It is trying to investigate whether formal state–business consultations (or simply PPD) and/or cronyism could counteract low institutional capacities and provide the needed mechanisms for fostering innovation. It is doing this with the help of a theoretical discussion followed by an empirical study based on an OLS multivariate panel econometric model. The novelty of this paper rests on trying to explore an important, yet under-researched, topic. There is insufficient research on how innovation is affected generally by SBR, and more specifically by either formal or informal forms of SBR. Empirical studies are even scarcer in this regard. These are questions that need to be explored in more depth, something that hardly exist in the present literature.

¹ PPD is an initiative that was boosted on a global level when an international workshop for PPD was organized in 2006 by the World Bank, UK Department for International Development (DFID) and the Organization for Economic Cooperation and Development (OECD), setting a charter for good practices aiming at private sector development (Herzberg & Wright, 2006, p. 10).

2 Theoretical background

Innovation, as argued above, is crucial for economic growth, raising the interest in exploring the important question of how innovation could be fostered? Three possible means to realize this objective could be thought of.

The first is to focus on direct government intervention to support innovation, for instance through subsidizing R&D, either using targeted or untargeted subsidies (see discussion in Aghion and Howitt 1998). The second is through supporting financial institutions targeting innovation. For example, such support would be in the form of policies and regulations encouraging the activities of Private Equity firms (PE), and especially Business Angels (BA) and Venture Capital (VC). PE have proved to be valuable for the growth of innovation worldwide (Alsina 2013, p. 24); and BA and VC are now widely involved in financing new business in the pre-seed, seed and early stages (see OECD 2015).

The third mean to foster innovation is through pursuing a more comprehensive mechanism that rests on the concept of National Innovation Systems (NIS). According to OECD (1997), the NIS framework regards technology and information flows among institutions, enterprises and individuals as crucial to innovation. According to the OECD (1997, p. 10), NIS refers to a network of institutions encompassing private and public sectors, whose interactions and activities result in developing, importing, modifying and spreading new technologies. The main inter-actors' "basic knowledge flows" include: Interactions among firms; interactions among firms, public research institutions and universities; spread of technology and knowledge to firms; and movement of personnel. As highlighted by Watkins et al. (2015), NIS places an emphasis on the central role of the government in promoting innovation. The partners of the government and its related agencies include: businesspeople; universities; and other private and public organizations engaged in educating activities. Recent works on NIS, according to the authors, focused more on industrial associations as representatives for businesspeople within the NIS. The importance of these associations is magnified in the case of developing countries where weak institutions are operating; in these countries, the interactions between industrial associations and the state should work on establishing the needed business environment for innovation. It is in this context that State Business Relations (SBR) could lead to the creation of a "growth coalition", argued Watkins et al. (2015, p. 1414). This could take the form of a "strong growth coalition" characterized by active state–business cooperation (Bräutigam et al. 2002; Watkins et al. 2015, p. 1414), sharing of information and having a high level of mutual "reciprocity, trust and credibility" (Schneider and Maxfield 1997; Watkins et al. 2015, p. 1414).

The various discussed-above approaches show that well-structured policies and regulations as well as favorable institutions are needed to realize a breakthrough in innovation. This is especially true with regard to the NIS approach. The need for coordination, cooperation, exchange of information and trust in order to have a successful NIS, all point to the need for proper institutions fostering these mechanisms and values. Such institutions could be grouped into two main categories: Governance-related institutions and SBR institutions. Other institutions, such as cultural and legal institutions, could also be important; but for the sake of

concision and simplicity, only the two previously mentioned institutional groupings would be investigated.

Starting with Governance-related institutions, Good Governance is argued in many works (e.g.: Keefer 2007) to lead to good economic outcomes. Realizing better innovational capacities should not arguably be exceptional in this regard. The Governance indicators as explained in Kaufmann et al. (2009) provide a valuable metric, although not fully comprehensive, for the formal institutional fabric of world countries. Among its six indicators, three are the most important for NIS: Regulatory quality; government effectiveness; and voice and accountability. A higher performance in regulatory quality should reflect higher government capability in conducting useful regulations that should benefit the private sector and its development, where innovation is important dimension in this regard. Higher government effectiveness would mean that state's bureaucracy and agencies are better equipped to implement various innovation-related support services, and are more capable of coordinating and dealing with the various partners within the NIS framework. Higher level of voice and accountability should mean the presence of the needed climate for the growth of various associations and civil society groups, and guarantee more strength to business associations, which should, as the literature on NIS suggests, lead to a better innovational system.

State–business relations (SBR), on the other hand, are also important, especially if wide institutional gaps exist with leading innovational and technological performers. State–business consultation should bring business people's input together with that of the government, where both parties' interaction is being shaped by the institutional settings in which both are operating. It is believed that SBR should cover for both sides' failures, something that poses their cooperation as being essential for economic development. On one hand, business failures refer to coordination failures in various aspects which affect technological development and others (Leftwich and te Velde 2010, pp. 6–7). Businesspeople could be unable to discover potentially profitable new products and would need public-provided research and development and legislations (Hausmann et al. 2008). Businesspeople may, moreover, suffer from scarcity of information and inability to efficiently use whatever information under their disposal (UNIDO 2000). On the hand, government failures include misallocation potential (Leftwich and te Velde 2010; te Velde 2009), but also imperfect information and foresight (Hausmann and Rodrik 2003). Furthermore, in developing countries the public sector often lack sufficient technological and administrative capacities as well as lack needed financial resources (UNIDO 2000).

By addressing each side's failure, SBR should affect innovational and technological capacities positively, arguably through promoting: information exchange; insightful policy recommendations; and building trust. Information exchange between businesspeople and state officials has been argued by Haggard et al. (1997) to lead to lowering transaction costs and improving policy design. Moreover, as Haggard et al. (1997, p. 55) argued, intensive information exchange resulting from participants' repeated interactions would foster trust and cooperation. These could be thought of as positive externalities which should positively affect political economic outcomes, with reference to the literature on social capital.²

² See Putnam et al. (1993), Haggard et al. (1997) and Rothstein and Stolle (2008).

State–business frequent interactions would develop connections and ultimately, as argued by Moon and Prasad (1994, p. 371), a richly bound network. According to the type of connections binding state officials and businesspeople, two forms of SBR could be differentiated: Informal and formal SBR. Connections resulting from familial, tribal or any social reason are likely to develop informal SBR between state officials and their connected businesspersons. The same is true for political connections, such as in the case when businesspersons are members in the ruling party or assume official positions. On the other hand, connections developing from taking part in an institutionalized and formal kind of state–business consultations—such as the ones to be found in economic councils or public private dialogues (PPD)—would result in formal SBR.³ The distinction between formal and informal SBR is important for investigating their (possibly) different effect on innovation.

Formal SBR is suggested to lead to good economic outcomes using the same three above-highlighted channels used generally by SBR: Information exchange (Campos and Roots 1996); improvement in public policy and planning (UNIDO 2000); and trust-building (Herzberg and Wright 2006, p. 12; Schneider 2010). The same channels should apply also for innovation. For instance, as discussed above, information exchange is very crucial for developing a well-functioning National Innovation System as well as for Private Equity VCs and BAs and their investment in innovation-led firms. Better information should also help governments in their R&D subsidy schemes. Network benefits resulting from formal SBR should also impact positively on the development of VCs and BAs. As suggested by OECD (2015, p. 80), private equity mechanisms need “a diverse network of knowledge producers, users and institutions” cooperating and exchanging knowledge. Moreover, networking between universities, research centers, businesspeople, VCs and BAs and the resulting trust and willingness to cooperate are what the process of developing innovational capacity needs.

Agreeing with the perspective of Crescenzi et al. (2013), formal SBR produce ‘bridging’ rather than ‘bonding’ social capital, that to say weakly connected but widely spread connections. To the contrary, informal SBR, especially when based on familial social connections, represent ‘bonding’ social capital given the intensity of the existing connections. As Crescenzi et al. (2013) argue and show empirically, bridging social capital is responsible for more innovation through active exchange of valuable “complementary knowledge”. On the other hand, bonding social capital results in “closed network” with “redundant knowledge” and “cognitive stagnation” and its effect on innovation is either insignificant or rather negative.

In terms of the National Innovation System approach, formal SBR (and especially councils and PPDs having various stakeholders) could foster the four presented aspects: interactions among firms, interactions among firms, public research institutions and universities; spread of technology and knowledge to firms; and movement of personnel. PPD could be the platform for inter-firms interactions and for interactions between firms on one hand and research and academic institutions on

³ For more on PPD and formal SBR, see Sabry (2013) and Sabry (2017B). For more on informal SBR, see Sabry (2017a).

the other. The latter interactions could indirectly encourage movement of personnel as a byproduct of networking and exchange of information and views. And, finally, the availability of more information and the interaction with VC managers and BAs with their great financing resources and expertise should facilitate the spread of technology and knowledge to participating firms.

Nevertheless, concerns on rent seeking raise some skepticism on the positive role of SBR, and even formal SBR on innovation. Depending on the literature on rent seeking (e.g.: Olson 1982), Business Organizations as interest groups could seek distributional gains and would accordingly try to influence government regulations and policies (Mueller 2003, pp. 554–558). Such rents that industrial associations could obtain would discourage inter-sectoral competition and innovation (Watkins et al. 2015, p. 1409).

On the other hand, cronyism could be thought of as an alternative for formal state–business consultation and PPD. Depending on the developmental experience of many countries in East Asia, a body of literature attributing a positive effect of cronyism on economic outcomes was being presented. This suggested that cronyism lower transaction costs, help in the circulation of better information, agreement enforcement, cost reduction, creation of more opportunities and enhancement of credibility, and eventually lead to economic growth (Kang 2003). If these claims are true, cronyism could serve as a mechanism for state–business consultation and exchange of information. Certainly some concerns could be raised about whether innovations resulting from such crony conditions would lead to better economic outcomes rather than merely benefiting state officials’ private interests (crony allocations to relatives, predation, etc.). Yet, both are likely to coincide; economic growth would increase chances for regime and officials’ survival (see Sabry 2018). Economic growth and survival, on the other hand, would increase resources and prolong officials’ access to those resources, both matching the officials’ private interests.

State–business consultation and exchange of information based on crony relations would be more likely the result of informal state–business interactions. That is to say, they would be mainly based on social and political connections rather than formalized, institutionalized and transparent consultations with Business Organizations and other broadly-represented business entities. In fact, it is argued that formal SBR, especially when it is broadly-participated by businesspeople, minimize cronyism. By enabling businesspeople to monitor government performance, formal SBR help in resisting various forms of corruption (Campos and Root 1996; Herzberg and Wright 2005), where cronyism is certainly one of these. Empirical evidence was given in Sabry (2017b) that formal SBR with high official representation and where widely-participated business associations exist would help in minimizing cronyism in many cultures.

Yet, cronyism is widely regarded by many scholars as having rather major negative effects on economic outcomes. Cronyism is suggested, for instance, to distort economic incentives, augment transaction costs and foster rent seeking (Kang 2003). Enderwick (2005, p. 120, 124) argued that cronyism could lead to misallocation of resources, since the state could protect inefficient industries for the sake of its business cronies. More relevant to innovation is the argument that by facing lax competitive pressures because of government regulatory protection, connected businesspersons would be less willing to engage in innovation and technological upgrading (Enderwick

2005, p. 125). They would be induced to spend less on innovation, but would devote more resources to building political ties, argued Desai (2007). As Aghion et al. (2001) has shown, lack of cost privileges in monopolistic competitive sectors forces firms operating under fierce competition to adopt new technologies for the sake of (temporary) cost reduction; this would ultimately result in more growth for the sector. However, less sectoral growth takes place when leading firms in these sectors entertain “exogenous cost advantages”, which are unmatched by their competitors; such advantages discourage pursuing technological upgrading as a mean for reducing costs (Diwan et al. 2014, pp. 23–24). If crony privileges are the source of these cost advantages (see Diwan et al. 2014, pp. 23–24), one can anticipate how cronyism could harm innovation and technological upgrading, besides undermining sectoral growth.

The question of whether PPD (or generally formal SBR) or cronyism could provide the needed mechanisms for fostering innovation is hence not theoretically settled. This is also true when we account for conditions where institutional deficiencies such as lower regulatory quality and government effectiveness are persistent. These are conditions which call for looking for other mechanisms to counteract these deficiencies and help in fostering innovation; and they are conditions for which the need for finding an answer for the posed question is more pressing.

Given the above discussion, one could hypothesize that SBR mechanisms, whether formal SBR (PPD) or cronyism should foster innovation. Their contribution would be especially felt in societies suffering from institutional shortcomings. And finally, because of concerns on cronyism and how it could also harm innovation, formal SBR are expected to benefit innovation more than cronyism.

3 Methodology

The previous theoretical discussion presented PPD and cronyism as possible alternative mechanisms which could counteract low institutional capacities and lead to more innovation. Different points of view were also presented, which placed much skepticism on the arguments made. It is the task of this section to test these claims empirically in order to contribute to the previous debate and give evidence in support to or against the various presented claims.

This is investigated using a number of panel multivariate regressions. In the first regression, the effect of business associations on innovation is tested. The second regression accounts for both the effect of business associations and PPD on innovation. The third, however, tests the effect of cronyism on innovation when both business associations and PPD are almost non-existent. Then in the fourth the effect of both, business associations and PPD, from one side, and cronyism from the other, on innovation is being tested. The other four regressions—regressions 5, 6, 7 and 8—are the same as the first four regressions, with only one exception which is incorporating a lagged dependent variable among the explanatory variables. Introducing regressions 5, 6, 7 and 8 serves two purposes. The first is a theoretical one which is accounting for the possibility of the presence of a dynamic rather than static model for innovation. In the fashion of Keele and Kelly (2005), this should account for the idea that present levels of innovation are the function of past levels as well as new

policies or institutional changes. Theoretically, this would imply that, other things remaining equal, a technologically or innovation-wise advanced country would likely advance even further as compared to less advanced countries. This would be an argument that runs against technological and economic growth convergence and in favor of divergence, making it a perspective worth accounting for. On the other hand, the statistical reasoning is to correct for autocorrelation, which as mentioned later would be a bigger concern for the first four regressions in comparison for the last four ones. Regression 1 corresponds to 5, regression 2 to 6, regression 3 to 7 and regression 4 to 8.

Beside the effect of business associations, PPD and cronyism, of particular importance is testing whether they could counteract the effect of bad institutions on innovation. Here these are identified as: low regulatory quality; and low government effectiveness, both of which are expected to negatively affect government capacities to conduct policies and regulations including those related to innovation. Other interactions accounted for are those between each of PPD and business association with another Governance variable, voice and accountability. This Governance variable is argued to be important for PPD and business association lobbying, which justifies its inclusion together with its interaction terms. Moreover, a number of control variables are included. The choice of these variables is motivated by trying to isolate the variables that affect innovation other than those related to SBR. This includes: (from the Global Competitiveness Indicators) quality of education; availability of research and training services; (and from the Governance Indicators): Government effectiveness; and regulatory quality. However, the important variables for this empirical study are: Business association participation; high official representation in PPD; interaction of business association and high official representation in PPD (the PPD proxy); cronyism; the interactions of each of business association, PPD proxy and cronyism with low regulatory quality and low government effectiveness; and the interactions of business association and the PPD proxy with voice and accountability. Regression 4, which includes all important variables, is formulated as follows:

$$\begin{aligned}
 (\text{Innov.})_{it} = & \alpha + \beta_1(\text{Bus. Assoc.})_{it} + \beta_2(\text{PPD High Off. Rep.})_{it} \\
 & + \beta_3(\text{Bus. Assoc.} \times \text{PPD High Off. Rep.})_{it} + \beta_4(\text{Crony.})_{it} \\
 & + \beta_5(\text{Low Reg. Quality})_{it} + \beta_6(\text{Low Gov. Effect})_{it} \\
 & + \beta_7(\text{Voice and Acc.})_{it} + \beta_8(\text{Bus. Assoc.} \times \text{Low Reg. Qual.})_{it} \\
 & + \beta_9(\text{Bus. Assoc.} \times \text{Low Gov. Effect})_{it} \\
 & + \beta_{10}(\text{Bus. Assoc.} \times \text{Voice and Acc.})_{it} \\
 & + \beta_{11}(\text{Bus. Assoc.} \times \text{PPD High Off. Rep.} \times \text{Low Reg. Qual.})_{it} \\
 & + \beta_{12}(\text{Bus. Assoc.} \times \text{PPD High Off. Rep.} \times \text{Low Gov. Effect})_{it} \\
 & + \beta_{13}(\text{Bus. Assoc.} \times \text{PPD High Off. Rep.} \times \text{Voice and Acc.})_{it} \\
 & + \beta_{14}(\text{Crony.} \times \text{Low Reg. Qual.})_{it} + \beta_{15}(\text{Crony.} \times \text{Low Gov. Effect})_{it} \\
 & + \beta_{16}(\text{Control})_{it},
 \end{aligned} \tag{1}$$

Table 1 The governance and global competitiveness indicators

Indicator	What it measures
Voice and accountability	“Captures perceptions of the extent to which a country’s citizens are able to participate in selecting their government, as well as freedom of expression, freedom of association and a free media”
Government effectiveness	“Captures perceptions of the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation and the credibility of the government’s commitment to such policies”
Regulatory quality	“Captures perceptions of the ability of the government to formulate and implement sound policies and regulations that permit and promote private sector development”
Favoritism in decisions of government officials	“to what extent do government officials show favoritism to well-connected firms and individuals when deciding upon policies and contracts?”
Quality of education	A composite of consisting of “quality of the educational system”, “quality of math and science education”, “quality of management schools”, and “internet access in schools”
Availability of research and training ^a	“How available are high-quality, professional training services?”
Innovation	A composite consisting of “capacity for innovation”, “quality of scientific research institutions”, “company spending on R&D”, “University–Industry collaboration in R&D”, “Government procurement of advanced technology products”, “Availability of scientists and engineers”, “PCT patent applications” and “intellectual property rights”

Source: World Bank Governance Indicators and The Global Competitiveness Report 2016–2017 (Schwab and Sala-i-Martin 2016). The definitions of the Governance and Global Competitiveness indicators are quoted or paraphrased from their respective websites

^aIt is to be noted that this indicator is reported differently in different occasions in the Global Competitiveness Report. It refers to question 5.07 which is leading to the indicator once called: “Local availability of specialized training services” and elsewhere called “Local availability of specialized research and training services”

where data on the different included variables covers the time (t) range between 1990 and 2016 for (i) 215 countries.

Regression 1 excludes PPD, cronyism and their interactions. In regression 2, cronyism and its interactions are dropped, while in regression 3 business association, high official representation in PPD and their interactions are dropped. In regressions 4, 5 and 6 lagged levels of innovation ($t-1$) are added as an independent variable.

The sources from which the variables (including control variables) are obtained and their description are listed in Table 1, while Table 2 presents their summary

Table 2 Summary statistics

Variable	Mean	Median	Minimum	Maximum	Std. dev.	Skewness	Ex. kurtosis
PPD high official representation	0.3950	0.0000	0.0000	1.0000	0.4891	0.4296	-1.8155
Business Associations participation	0.5488	0.5502	0.0805	1.0000	0.2135	0.0400	-0.6328
Quality of education	3.9779	3.9002	1.7088	6.1711	0.9316	0.2271	-0.6471
Availability of research and training	4.0921	4.0155	1.9207	6.4978	0.9049	0.3081	-0.4530
Innovation	3.3953	3.1518	1.6762	5.8381	0.8583	1.0533	0.4375
Government Effectiveness	-0.0076	-0.1676	-2.4871	2.4313	1.0012	0.3291	-0.6812
Regulatory Quality	-0.0077	-0.1151	-2.6754	2.2629	0.9998	0.0152	-0.6757
Voice and Accountability	-0.0085	0.0104	-2.2862	1.8264	0.9999	-0.1914	-1.0096
GDP per capita growth annual	2.0394	2.1048	-64.9963	140.5010	6.2511	2.6487	78.5482
Low Government Effectiveness	0.5649	1.0000	0.0000	1.0000	0.4958	-0.2619	-1.9314
Low Regulatory Quality	0.5410	1.0000	0.0000	1.0000	0.4984	-0.1644	-1.9730
Bus. Ass. × Voice & Acc.	0.0091	-0.0317	-1.3046	1.2416	0.4206	0.3177	0.2728
Bus. Ass. × Low Gov. Effect.	0.1419	0.0000	0.0000	1.0000	0.2637	1.6456	1.3517
Bus. Ass. × Low Reg. Qual.	0.1254	0.0000	0.0000	1.0000	0.2511	1.8332	2.0386
Cronyism	4.7533	4.9680	1.9531	6.5197	0.9055	-0.8147	0.0772
Crony × Low Gov. Effect.	1.5055	0.0000	0.0000	6.5197	2.3946	0.9904	-0.9590
Crony × Low Reg. Qual.	1.3757	0.0000	0.0000	6.5197	2.3214	1.1270	-0.6603
PPD High Official Rep. × Bus. Ass.	0.1700	0.0000	0.0000	1.0000	0.2840	1.3232	0.2056
PPD High Official Rep. × Bus. Ass. × Voice & Acc.	0.0089	0.0000	-0.9587	1.0908	0.2482	0.5567	5.2036
PPD High Official Rep. × Bus. Ass. × Low Gov. Effect.	0.0444	0.0000	0.0000	1.0000	0.1613	3.7023	12.7483
PPD High Official Rep. × Bus. Ass. × Low Reg. Qual.	0.0372	0.0000	0.0000	1.0000	0.1473	4.0422	15.3981

statistics. The dependent variable, innovation, is obtained from the Global Competitiveness Indicators (GCI). It is a composite of different indicators including: “capacity for innovation”, “quality of scientific research institutions”, “company spending on R&D”, “university–industry collaboration in R&D”, “government procurement of advanced technology products”, “availability of scientists and engineers”, “PCT patent applications” and “intellectual property rights”. Such a comprehensive perspective for measuring innovation is what encouraged its use in this research. While other works in the literature used much limited proxies for innovation, such as growth rate of patents (see Crescenzi et al. 2013), the use of the GCI innovation indicator allows a better measurement of different aspects related to innovation. Moreover, it accounts for aspects of NIS (e.g.: university–industry collaboration) in each country. Data for this indicator is available for 140 countries from year 2007 until 2015.

Most of the independent variables are obtained from the GCI and Governance Indicators. The main independent variables of interest are, however, obtained as follows.

Cronyism is obtained from the opposite value of “favoritism in decisions of government officials”, one of the Global Competitiveness indicators (see Table 1). This indicator runs from 1 for high favoritism to 7 for low favoritism. Thus, for constructing the cronyism proxy, 8 is subtracted from each value to flip the indicator, making higher value (i.e.: the value of 7) indicating higher favoritism in decisions of government officials, or in other words cronyism.

As for business association participation, this proxy is obtained from the World Bank’s “Enterprise Surveys”, where businesspersons from different countries were asked whether they are or aren’t members of a business association. The percentage of those responding with a yes out of the total surveyed businesspersons in each country reflects to some extent the percentage of business people’s participation in business associations. Worth noting is that the proxy doesn’t reflect how fragmented or not businesspeople are. For instance, a higher business participation in business association could still mean that businesspeople are scattered among numerous and probably conflicting business associations. These business associations could also be controlled and manipulated by the government. Despite of these shortcomings, all things being the same, higher participation in business associations should more likely reflect business people’s ability to aggregate their power more than in the case where lower levels of participation exists.

PPD high official representation, on the other hand, is obtained as introduced in Sabry (2013). A brief summary is provided below. PPD, or more generally formal state–business consultation experiences, considered here are:

- those focusing on overall economic activity, the whole manufacturing or the whole services sectors, or many manufacturing and services sub-sectors.
- those of national character (not with provincial, regional, or international character).
- those characterized by formality and sustainability.

It should be noted that the considered PPD are not necessarily focusing on innovation, but their regulations and policy recommendations would in a way or another affect innovation. For instance, many of the considered PPD target business climate reforms. Easing regulation of entry is one of the pillars of such reforms; but easy entry and exit is also argued to be an innovation-fostering mechanism based on Schumpeterian “creative destruction” models (see Aghion and Howitt 1998).

For these considered PPD, high official representation is calculated as a dummy variable. High official representation (by a president, king, prime minister, minister or deputy minister) is given the value of 1. Otherwise, the value of the dummy variable is 0.

The interaction term between business association and high official representation in PPD is referred to as the PPD proxy. It is almost always the case that business representatives in the considered PPD are members of business associations. So, knowing the level of business participation in business associations is more likely to reflect how represented businesspeople are in the considered PPD. Thus, higher values of the interaction term, or the PPD proxy, should reflect both high official representation and high business participation. The collected data comprises PPD experiences and/or business association participation information of 115 countries for the time extending between 1990 and 2011.

Two more independent variables of interest are being calculated; these are low government effectiveness and low regulatory quality. Introducing these variables are of high relevance, since this study has among its objectives finding out whether SBR could counteract the effect of low institutional capacities, where these two Governance-related institutions are of high relevance to policy formulation and conduction. Both of these two variables are calculated as dummies, which are given the value of 1 when a country’s performance is below 0, and given the value of 0 otherwise (note that the values of the Governance Indicators always range between +2.5 for best performance and −2.5 for worst).

The used variables, whether control or main independent variables, raise arguably less theoretical concern with regard to endogeneity (specifically reverse causality). It is theoretically less likely that innovation affects either Governance indicators or PPD and cronyism rather than the other way round.

The conducted regressions are reported in Table 3. The General to Specific method is used to remove insignificant variables without significantly altering the conducted regressions. This is conducted automatically by the Gretl program used for running the regressions. It is performed through the program’s “sequential elimination at 10%” function; and an F-Test is provided to ensure the consistency of the method. The results of these tests are provided in Table 4.

The overall effect of important independent variables accounting for their interaction terms are shown in Table 5. To calculate the values presented in this table, the following equation is used:

$$\text{Total effect of } X \text{ on } Y = \beta_1 + \beta_2 Z, \quad (2)$$

where β_1 is the coefficient of X taken individually without interaction, and β_2 is the coefficient of the interaction term in which X is part of, and Z is the other variable of the interaction term. The value used for Z is that of its mean (see Table 2). Finally,

Table 3 The conducted regressions

	Dependent variable: innovation							
	Reg (1)	Reg (2)	Reg (3)	Reg (4)	Reg (5)	Reg (6)	Reg (7)	Reg (8)
Constant	0.728*** (0.141)	0.393** (0.177)	1.688*** (0.121)	2.106*** (0.314)	-0.096* (0.058)	-0.250*** (0.079)	0.281*** (0.054)	0.584*** (0.169)
Innovation (t-1)					0.810*** (0.027)	0.800*** (0.029)	0.842*** (0.011)	0.756*** (0.029)
Quality of education	0.219*** (0.035)	0.306*** (0.043)	0.058*** (0.022)	0.303*** (0.038)	0.085*** (0.019)	0.128*** (0.022)	0.039*** (0.009)	0.116*** (0.021)
Availability of research and training	0.448*** (0.031)	0.422*** (0.039)	0.547*** (0.019)	0.412*** (0.038)	0.088*** (0.021)	0.101*** (0.025)	0.086*** (0.010)	0.105*** (0.023)
Government Effectiveness	0.304*** (0.050)	0.191** (0.076)	0.299*** (0.036)					
Regulatory Quality	-0.126** (0.057)	-0.228*** (0.076)	-0.052* (0.031)	-0.219*** (0.068)		-0.111*** (0.022)		-0.115*** (0.024)
Voice and Accountability	-0.486*** (0.071)	-0.245*** (0.069)		-0.455*** (0.083)	-0.053** (0.026)			
Low Gov. Effect.		0.466*** (0.152)		-1.056*** (0.332)				-0.574*** (0.174)
Low Reg. Qual.	-0.247* (0.131)	-0.570*** (0.168)		-0.666*** (0.166)	0.059*** (0.021)			
Business Association participation	-0.262*** (0.122)			-0.672*** (0.134)				-0.069* (0.036)
Bus. Ass. × Voice & Acc.	0.805*** (0.130)	0.323** (0.124)		0.779*** (0.154)	0.093** (0.045)	0.048 (0.030)		0.048* (0.028)
Bus. Ass. × Low Gov. Effect.		-1.013*** (0.269)						
Bus. Ass. × Low Reg. Qual.	0.611*** (0.204)	1.061*** (0.262)		1.098*** (0.252)				

Table 3 (continued)

Dependent variable: innovation								
	Reg (1)	Reg (2)	Reg (3)	Reg (4)	Reg (5)	Reg (6)	Reg (7)	Reg (8)
PPD high official representation		0.204*** (0.077)		0.117*** (0.032)		0.041** (0.018)		0.040** (0.018)
PPD		−0.305** (0.132)						
PPD × Voice & Acc.		0.301*** (0.093)		0.134* (0.075)				
PPD × Low Gov. Effect.		0.339** (0.130)						
Cronyism			−0.192*** (0.015)	−0.242*** (0.044)			−0.052*** (0.007)	−0.124*** (0.022)
Crony × Low Gov. Effect.			0.015** (0.007)	0.179*** (0.063)				0.111*** (0.033)
Crony × Low Reg. Qual.			0.029*** (0.007)				0.005*** (0.002)	
N	347	243	1229	243	275	196	1074	196
R-squared	0.82	0.81	0.87	0.82	0.96	0.95	0.98	0.96
Adjusted R-squared	0.82	0.80	0.87	0.82	0.95	0.95	0.98	0.96
Durbin–Watson	0.18	0.21	0.17	0.25	1.14	1.04	1.33	1.15

Table 4 Tests done according to the general to specific method

	Eliminated variables (sequential elimination using two-sided $\alpha=0.10$)	F test
Reg. 1	Low Gov. Effect, Bus. Assoc. $\times$ Low Gov. Effect	$F(2, 335)=0.773217$
Reg. 2	Business Association Particip., PPD $\times$ Low Reg. Qual.	$F(2, 226)=0.310073$
Reg. 3	Voice and Account., Low Gov. Effect, Low Reg. Qual.	$F(3, 1218)=0.719454$
Reg. 4	Gov. Effect, Bus. Assoc. $\times$ Low Gov. Effect, PPD, PPD $\times$ Low Gov. Effect, PPD $\times$ Low Reg. Qual., Crony $\times$ Low Reg. Qual.	$F(6, 223)=0.664757$
Reg. 5	Gov. Effect, Regul., Low Gov. Effect, Bus. Assoc. Particip., Bus. Assoc. $\times$ Low Gov. Effect, Bus. Assoc. $\times$ Low Reg. Qual.	$F(6, 262)=0.964431$
Reg. 6	Step 1: Gov. Effect, Low Reg. Qual., Bus. Assoc. $\times$ Low Reg. Qual., PPD, PPD $\times$ Voice. and Acc., PPD $\times$ Low Gov. Effect, PPD $\times$ Low Reg. Qual.	Step 1: $F(7, 178)=0.207312$ Step 2: $F(4, 185)=1.29011$
Reg. 7	Step 2: Voice and Acc., Low Gov. Effect, Bus. Assoc. Particip., Bus. Assoc. $\times$ Low Gov. Effect	
Reg. 8	Gov. Effect, Reg. Qual., Voice and Acc., Low Gov. Effect, Low Reg. Qual., Crony $\times$ Low Gov. Effect	$F(6, 1062)=0.690725$
	Gov. Effect, Voice and Acc., Low Reg. Qual., Bus. Assoc. $\times$ Low Gov. Effect, Bus. Assoc. $\times$ Low Reg. Qual., PPD, PPD $\times$ Voice and Acc., PPD $\times$ Low Gov. Effect, PPD $\times$ Low Reg. Qual., Crony $\times$ Low Reg. Qual.	$F(10, 175)=0.640974$
Reg. A1	Gov. Effect, Reg. Qual., Low Reg. Qual., Bus. Assoc. Particip., Bus. Assoc. $\times$ Low Gov. Effect, Bus. Assoc. $\times$ low Reg. Qual., PPD, PPD $\times$ Voice and Acc., PPD $\times$ Low Gov. Effect, PPD $\times$ Low Reg. Qual., Crony $\times$ low Reg. Qual.	$F(11, 220)=1.039$
Reg. A2	Gov. Effect, Voice and Acc., Low Reg. Qual., Bus. Assoc. $\times$ Low Reg. Qual., PPD, PPD $\times$ Low Gov. Effect, Crony $\times$ Low Reg. Qual.	$F(7, 172)=0.518474$
Reg. B1	Reg. Qual., Bus. Assoc. $\times$ Voice and Acc., Bus. Assoc. $\times$ Low Reg. Qual., PPD $\times$ Low Reg. Qual.	$F(4, 170)=0.133089$
Reg. B2	Voice and Acc., Bus. Assoc. Particip., Bus. Assoc. $\times$ Low Gov. Effect, PPD	$F(4, 125)=0.105697$
Reg. B3	Gov. Effect, Bus. Assoc. Particip., Bus. Assoc. $\times$ Voice and Acc., Bus. Assoc. $\times$ Low Gov. Effect	$F(4, 167)=0.054026$
Reg. B4	Gov. Effect, Voice and Acc., Bus. Assoc., PPD, Cronyism	$F(5, 122)=0.307623$

Table 5 Overall effect of variables in comparison to their effect without their interactions

	Reg 1		Reg 2		Reg 3		Reg 4		Reg 5		Reg 6		Reg 7		Reg 8	
	Without	Overall	Without	Overall	Without	Overall	Without	Overall	Without	Overall	Without	Overall	Without	Overall	Without	Overall
Business Association	-0.262	0.342	-	-0.260	-	-	-0.672	0.418	-	-0.001	-	-	-	-	-0.069	-0.069
PPD High Gov. Repres.	-	-	0.204	0.087	-	-	0.117	0.118	-	-	0.041	0.041	-	-	0.040	0.040
PPD High gov. Rep. × Bus. Ass.	-	-	-0.305	-0.116	-	-	-	-0.001	-	-	-	-	-	-	-	-
Cronyism	-	-	-	-	-0.192	-0.148	-0.242	-0.063	-	-	-	-	-0.052	-0.047	-0.124	-0.013
Voice and Account.	-0.486	-0.044	-0.245	-0.017	-	-	-0.455	-0.005	-0.053	-0.002	-	-	-	-	-	0.026
Low Gov. Effect	-	-	0.466	-0.032	-	-	-1.056	-1.056	-	-	-	-	-	-	-0.574	-0.046
Low Reg. Qual.	-0.247	0.088	-0.570	0.012	-	-	-0.666	-0.063	0.059	0.059	-	-	-	0.024	-	-

Standardized Beta and economic significance are presented in Table 6. For Standardized Beta the following equation is used:

$$\text{Standardized } \beta_i = \beta_i [\sigma(x_i) / \sigma(y)], \quad (3)$$

Where $\sigma(x_i)$ and $\sigma(y)$ are standard deviations of the independent and dependent variables respectively. Economic significance is measured in a fashion that shows the percentage change in independent variables needed to induce a 1% change in the dependent variable. The calculation is derived from the following equation measuring relative percentage change:

$$\% \Delta Y / \% \Delta X_i = [\Delta Y / \text{Range } Y] / [\Delta X_i / \text{Range } X_i], \quad (4)$$

Where the range of each of X and Y is the difference between the maximum and minimum values of each of them. From Eq. (4), the following formula is derived:

$$\% \Delta X_i = [1 / \beta_i] [\text{Range } Y / \text{Range } X_i] [\% \Delta Y], \quad (5)$$

where $\% \Delta Y$ is being replaced by the value of 1 to reflect 1% change in the dependent variable (i.e.: innovation).

4 Results and discussion

4.1 The main regressions

Looking at the results reported in Table 3, it is to be noted that the first four conducted regressions have managed to explain much of the variability of the dependent variable (innovation). The adjusted R-squared of the four regressions are ranging between 0.8 and 0.87. The Durbin Watson coefficients, however, raise concerns on positive serial autocorrelation. As mentioned before, this is what partly induced conducting regressions 5, 6, 7 and 8 having a lagged dependent variable, innovation ($t - 1$), being accounted for among the explanatory factors. In these three regressions higher Durbin Watson coefficients are realized indicating significant improvement in reducing positive autocorrelation, yet without totally eradicating it.⁴ These three regressions have also better fit as reflected in their R-Square and Adjusted R-Square values.

In the following lines, only statistically significant variables are reported. Most of the included control variables show the expected sign with regard to their effect on innovation. The two included Global Competitiveness variables, quality of education and availability of research and training, affect innovation positively in all of the conducted regressions. The three included Governance variables have, however, different effects. Government effectiveness showed the expected positive effect on innovation in regressions 1, 2 and 3. Regulatory quality, on the other hand, have

⁴ It is to be noted that it is generally suggested that Durbin Watson should not be used in measuring autocorrelation in models having lagged dependent variables. However, in the literature it is also argued that Durbin's H (the suggested test for autocorrelation in this case) and the standard Durbin Watson coefficient (d) would roughly give the same decision with regard to autocorrelation even when lagged dependent variables are included (see for instance Sapir, 1977).

Table 6 Economic significance and standardized beta

Variable	Percentage change in independent variable needed to cause 1% change in innovation								Standardized beta							
	Reg1%	Reg2%	Reg3%	Reg4%	Reg5%	Reg6%	Reg7%	Reg8%	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5	Reg 6	Reg 7	Reg 8
Bus. Assoc. Partic.	-22.901	-	-	-8.929	-	-	-	-86.957	-0.065	-	-	-0.167	-	-	-	-0.017
Bus. Assoc. × Voice Acc.	1.491	3.715	-	1.540	12.903	-	-	25.000	0.394	0.158	-	0.382	0.046	-	-	0.024
Bus Assoc. × Low Gov. Effect	-	-5.923	-	-	-	-	-	-	-	-0.311	-	-	-	-	-	-
Bus Assoc. × Low Reg. Qual.	9.820	5.655	-	5.464	-	-	-	-	0.179	0.310	-	0.321	-	-	-	-
PPD high off. repres.	-	-	-	-	-	-	-	-	-	0.116	-	0.067	-	0.023	-	0.023
PPD high off. rep. × Bus. Ass.	-	-19.672	-	-	-	-	-	-	-	-0.101	-	-	-	-	-	-
PPD high gov. rep. × Bus. Ass. × Voice Acc.	-	3.987	-	8.955	-	-	-	-	-	0.087	-	0.039	-	-	-	-
PPD high off. rep. × Bus. Ass. × Low Gov. Effect	-	17.699	-	-	-	-	-	-	-	0.064	-	-	-	-	-	-
PPD high gov. rep. × Bus. Ass. × Low Reg. Qual	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 6 (continued)

Variable	Percentage change in independent variable needed to cause 1% change in innovation								Standardized beta							
	Reg1 %	Reg2 %	Reg3 %	Reg4 %	Reg5 %	Reg6 %	Reg7 %	Reg8 %	Reg 1	Reg 2	Reg 3	Reg 4	Reg 5	Reg 6	Reg 7	Reg 8
Cronyism	-	-	-5.208	-4.132	-	-	-19.231	-8.065	-	-	-0.203	-0.255	-	-	-0.055	-0.131
Crony × Low Gov. Effect	-	-	66.667	5.587	-	-	-	9.009	-	-	0.042	0.499	-	-	-	0.310
Crony × Low Reg. Qual.	-	-	34.483	-	-	-	200.000	-	-	-	0.078	-	-	-	0.014	-

negative effects except in regression 5 and 7; and voice and accountability has a negative effect in regressions 1, 2, 4 and 5.

Focusing on the first four regressions (in which lagged innovation was not included) and on independent variables of interest, one notices the following. High business association participation tends to affect innovation positively, and this could be detected from its overall effect in both regressions 1 and 4 (although the sign is different in regression 2) as shown in Table 5 (and this accounts also for the interactions of the PPD proxy which is in itself a common interaction term of PPD high official representation and business association). The variables' positive effect is reinforced by the presence of higher levels of voice and accountability, as the results for regressions 1, 2 and 4 show. High business association participation seems to succeed in fixing the effect of low regulatory quality and lead to higher levels of innovation, again in regressions 1, 2 and 4. It is to be noted that low regulatory quality on its own, that is to say without interaction with business association, has a negative effect on innovation in the three mentioned regressions. Yet, the interaction of the variable with business association has a positive effect on innovation. Nevertheless, business association participation doesn't succeed in fixing the effect of low government effectiveness, where their interaction is negative in regression 2 and insignificant elsewhere.

Surprisingly, when high levels of business association participation coexist with high official representation in PPD the results seem not to be in favor of innovation, as the overall effect of this interaction term reveals (see Table 5). This is despite the observation that high official representation in PPD on its own (without interaction) has an overall positive effect in both regressions 2 and 4. The interaction term (the PPD proxy) however, leads to higher innovation when high levels of voice and accountability exist. This is evident from the results of the interaction term of PPD proxy and voice and accountability in both regressions. The PPD proxy also manages to fix the effect of low government effectiveness and lead to higher innovation. But, this is only true when cronyism is non-existent (Regression 2). When cronyism is present (in Regression 4), it seems to lose this advantage to the favor of the interaction term between cronyism and low government effectiveness.

According to Table 5, in both the third and the fourth regressions in which cronyism is present, the overall effect of cronyism is negative on innovation. In both regressions, cronyism without interactions has also a negative effect. Nevertheless, cronyism interaction with low government effectiveness leads to high innovation in both regressions. Its interaction with low regulatory quality also improves innovation, but only in regression 3, and becomes insignificant in regression 3 when high business association participation exists. In fact, it seems cronyism loses the advantage of fixing low regulatory quality to business association's favor in that regression (regression 4).

Moving to the regressions where lagged dependent variable is included (regressions 5, 6, 7 and 8), one can notice that many variables which were significant in the corresponding regressions (regression 1 versus 5, regression 2 versus 6, regression 3 versus 7 and regression 4 versus 8) became insignificant and were removed by the General to Specific method. However, for the remaining variables the sign was maintained in almost every case. The lagged innovation variable is significant

in the four regressions and has a positive effect. The interaction between business association and voice and accountability remains significant in both regression 5 and 8. High official representation in PPD remains significant and have a positive effect in regressions 6 and 8 (the same as it was in the corresponding regressions of 2 and 4 but with a lower slope). As for cronyism, it remains with a negative effect on innovation when taken without its interactions. It still counteracts low regulatory quality in regression 7 (as in regression 3) and low government effectiveness in regression 8 (as in regression 4) and leads to higher innovation.

How could these results be understood? The first question to tackle is whether more credit should be given to the first four regressions or the last four ones? It is true that the last four regressions have a better fit and less autocorrelation, making them look more credible. However, accounting for the perspective introduced in Achen (2000), one should question this assumption. The introduction of the lagged dependent variable might have suppressed independent variables that were significant when it was not included, without this necessarily being theoretically justified. It all rests on which theoretical assumptions one would take. In this case, the decision is between the assumption of technological and growth divergence or lack of such tendency.

Assuming that there is no tendency for divergence gives more credit for the first four regressions (despite the problem of higher autocorrelation). According to these, the obtained results would make one conclude that cronyism harms innovation. Yet, at low government effectiveness, low regulatory quality, weak or non-existent business associations and non-existent or lower officially-represented PPD, cronyism seems to offer a remedy. In this case favorite allocation of resources seems to counteract the effect of low government effectiveness and low regulatory quality and lead to more innovation. Even when broadly-participated business associations and highly officially represented PPD both exist, cronyism could still counteract the effect of low government effectiveness. In fact, cronyism does so at the time where the other two variables fail to. But, this is not true for low regulatory quality. There is empirical evidence that at this case, high levels of participation in business association would be the variable counteracting this institutional deficiency, leading ultimately to higher innovation. And what further reinforces this belief in business association participation is that it manages to counteract low regulatory quality in the three regressions in which it exists, whether there is cronyism or not.

Another important suggestion is that voice and accountability is very important to unleash the positive effect on innovation of both high business association participation and the interaction of this variable with high Official representation in PPD (PPD proxy). Furthermore, high official representation in PPD is important for innovation, especially at lower levels of business participation in business association, where these associations are supposed to be business partners for the government in these PPD. It seems that when Businesspeople fail to aggregate their power, the need for government high representation increases, probably for coordination purposes and the high need for government resources (including information). On the other hand, noticing that the coexistence of high officially-represented PPD and broadly-participated business associations succeed in counteracting low government effectiveness, but only when cronyism is eradicated, suggests that fighting cronyism is a

mean for promoting the role of highly officially-represented and broadly business-participated PPD in fixing institutional deficiencies and leading to more innovation.

Taking the assumption of divergence and putting the last four regressions into account would give a different but not a much contradicting view. In this perspective, cronyism still harms innovation, but still counteracts institutional deficiencies of low government effectiveness and low regulatory quality, managing to fix the former when neither PPD nor business association succeed in doing so. Voice and accountability remains important for helping high participation in business associations in leading to higher innovation. However, any effect of either PPD or business association in counteracting institutional deficiencies disappears.

With the help of Table 5, one can also note that the overall effect of cronyism on innovation in regression 7 is lower than in regression 3 (-0.047 in comparison to -0.148). The same is true between regressions 8 and 4 (-0.013 in comparison to -0.063). This suggests that in the regressions assuming divergence (regressions 7 and 8), the effect of cronyism on the incremental changes in innovation is much lower than when we don't assume divergence (regressions 3 and 4). In other words, accounting for initial levels of innovation as an independent variable shows that present levels of cronyism are less economically significant in decreasing innovation. Conversely, removing initial innovation conditions from the regression makes the effect of current levels of cronyism appear much larger on present levels of innovation. The above remark implies that current levels of cronyism correlate with countries' different initial levels of innovation (because adding initial levels of innovation decrease the effect of cronyism on the dependent variable). Moreover, the results show that cronyism still correlates with current levels of innovation. On the other hand, it is to be noted that changes in cronyism between two successive years are likely to be minimal; that is to say, it is highly likely that current levels of cronyism greatly reflect its past levels. Hence the correlation between current levels of cronyism and initial levels of innovation could be in fact reflecting the correlation between past levels of cronyism and initial levels of innovation. These initial levels of innovation themselves could have evolved through time, partially because of cronyism. Thus, cronyism probably explains past levels of innovation the same as it explains incremental decreases in innovation. In this sense, it could be said that cronyism is probably one of the reasons for divergence in terms of innovation, and ultimately in terms of economic growth.

4.2 Robustness

Having done the previous analysis, two things call for further investigation. The first is whether the used indicator for innovation is the best possible proxy for representing the dependent variable. As described above, the used indicator for innovation obtained from the GCI is a composite having different variables. It is to be noticed that some of these variables are likely unrelated to SBR, PPD or cronyism. This is the case, for example, for "availability of scientists and engineers". The second concern is whether the fixed effects regression model should be used instead of OLS for the conducted regressions. There are possibly missing time-invariant cultural

variables unaccounted for in the OLS regressions conducted in the previous sub-section. There are also probably common historical events of a global nature that have affected innovation in a given year. Both kinds of missing variables are most probably correlated with the already used independent variables, which calls for the use of a fixed effects, rather than a random effects, model. Yet, as a sign of caution, fixed effects model isn't usually much helpful when the variables of concern are either time invariant or with little time invariability. In the dataset used in this work, business association participation, high official representation in PPD and the PPD proxy (the interaction between the previous two variables) are all low or non-time invariant in each studied country. Despite of these precautions, the use of fixed effects model is still worthwhile.

Given the above-mentioned concerns, the robustness of the results obtained in the main regressions sub-section are investigated here by two sets of new regressions. The results of these regressions are reported in Tables 7 and 8. For simplicity, the formula having all independent variables of interest (Business Association, PPD, cronyism and their interactions) used in regressions 4 and 8 is going to be used instead of having to do once again step by step addition of independent variables.

The first set of regressions, regressions A1 and A2, are again OLS regressions having the same dependent and independent variables as regressions 4 and 8 respectively; thus, regression A2 has a lagged innovation variable among the independent variables, while A1 doesn't. The only exception between regression A1 and regression 4 and between regression A2 and regression 8 is that more control variables are now added to the new regressions (A1 and A2). These control variables are those which constitute the innovation composite variable, but are likely less affected by SBR variables. By isolating their effect, the remaining part of the variation in the innovation composite is more likely to be affected and related to SBR. Accordingly, from the variables constituting the innovation composite, three variables are placed as control variables in the new regressions. These are: "quality of scientific research institutions", "University–Industry collaboration in R&D", and "Availability of scientists and engineers". The rest of the variables constituting the innovation composite are more likely to be affected by SBR directly or indirectly; and hence these are not controlled for.

The second set of regressions are all conducted by the fixed effects regression model with both country and time dummies. These include regressions B1 and B2 corresponding to regressions 4 and 8 (respectively) of the main regressions, with B2 having a lagged innovation variable among the independent variables. It also includes regressions B3 and B4 corresponding to regressions A1 and A2 respectively, where more control variables are added (from among the variables constituting the innovation composite) and where B4 has a lagged innovation variable among the independent variables.

The results in Table 7 show that adding the above-mentioned control variables improved the fit, with both higher R-squared and adjusted R-squared for regressions A1 and A2 in comparison to regressions 4 and 8 respectively. On the other hand, the results of A1 and A2 didn't generally much offshoot in comparison to the main regressions. As general comments on statistically significant effects, one can say the following about the obtained results. Voice and accountability helps

Table 7 OLS regressions for investigating robustness

Pooled OLS: dependent variable: innovation

	Reg A1	Reg A2
Constant	0.764*** (0.194)	0.517*** (0.158)
Innovation (t – 1)	NA	0.571*** (0.038)
Quality of education	0.071*** (0.024)	0.084*** (0.019)
Availability of research and training	0.104*** (0.031)	0.052** (0.024)
Quality of scientific research	0.238*** (0.026)	0.082*** (0.023)
University industry collaboration	0.237*** (0.032)	0.105*** (0.027)
Availability of scientists and engineers	0.162*** (0.020)	0.076*** (0.017)
Government Effectiveness		
Regulatory Quality		– 0.114*** (0.025)
Voice and Accountability	– 0.130*** (0.030)	
Low. Gov. Effect.	– 0.502** (0.198)	– 0.610*** (0.171)
Low Reg. Qual.		
Business Association participation		– 0.095* (0.054)
Bus. Ass. × Voice & Acc.	0.188*** (0.052)	0.082** (0.037)
Bus. Ass. × Low Gov. Effect.		0.106 (0.078)
Bus. Ass. × Low Reg. Qual.		
PPD high official representation	0.053** (0.021)	0.066*** (0.021)
PPD		
PPD × Voice & Acc.		– 0.073 (0.046)
PPD × Low Gov. Effect.		
PPD × Low Reg. Qual.		– 0.096** (0.046)
Cronyism	– 0.127*** (0.027)	– 0.121*** (0.021)
Crony × Low Gov. Effect.	0.099*** (0.038)	0.110*** (0.030)
Crony × Low Reg. Qual.		
N	243	196
R-squared	0.927	0.969
Adjusted R-squared	0.923	0.966
Durbin–Watson	0.153	0.877

business associations in leading to higher innovation, high official representation in PPD increases innovation and PPD fail in overcoming lower regulatory quality with regard to its effect on innovation. The overall effect of cronyism on innovation is negative;⁵ but it manages to overcome low government effectiveness leading in this specific case to higher innovation.

Regressions B1, B2, B3 and B4, however, don't provide enough helpful information, since most of the variables of interest are statistically insignificant, and those

⁵ In regression A1, the overall effect of cronyism = $-0.127 + (1 \times 0.099) = -0.028$ and in regression A2 = $-0.121 + (1 \times 0.110) = -0.011$

Table 8 Fixed effects regression for investigating robustness

Fixed-effects model: dependent variable: innovation				
	Reg 1	Reg 2	Reg 3	Reg 4
Constant	0.057*** (0.014)	0.052*** (0.013)	0.025** (0.010)	0.033*** (0.009)
Innovation (t - 1)		-0.132** (0.060)		-0.054 (0.038)
Quality of education	1.590*** (0.294)	1.675*** (0.382)	0.601*** (0.220)	0.473** (0.226)
Availability of research and training	0.330*** (0.034)	0.340*** (0.037)	0.192*** (0.025)	0.206*** (0.025)
Quality of scientific research			0.097*** (0.020)	0.096*** (0.020)
University industry collaboration			0.180*** (0.021)	0.129*** (0.021)
Availability of scientists and engineers			0.166*** (0.026)	0.194*** (0.024)
Government Effectiveness	0.198*** (0.029)	0.177*** (0.031)		
Regulatory Quality		-0.103 (0.096)	0.112*** (0.019)	0.114*** (0.021)
Voice and Accountability	-0.044 (0.072)		0.091** (0.042)	
Low Gov. Effect.	-0.239* (0.126)	0.052 (0.080)	-0.174* (0.105)	0.123** (0.049)
Low Reg. Qual.	-0.944*** (0.310)		-0.280 (0.185)	
Business Association participation	1.226*** (0.289)			
Bus. Ass. × Voice & Acc.		-0.883*** (0.264)		-0.320 (0.205)
Bus. Ass. × Low Gov. Effect.	-0.638 (0.573)			-0.316 (0.234)
Bus. Ass. × Low Reg. Qual.		-0.499 (0.369)	0.290 (0.302)	0.152 (0.315)
PPD high official representation				
PPD	-0.566 (0.470)		-0.973 (0.741)	
PPD × Voice & Acc.	1.138 (0.754)	-1.946 (1.417)	0.554 (0.359)	-1.740* (0.964)
PPD × Low Gov. Effect.	-0.446* (0.250)	-0.363 (0.414)	-0.142 (0.196)	-0.130 (0.263)
PPD × Low Reg. Qual.		-0.597* (0.318)	-0.926** (0.408)	-0.421 (0.368)
Cronyism	-0.539 (0.790)	1.756 (1.481)	1.173 (0.823)	
Crony × Low Gov. Effect.	-0.055* (0.029)	-0.046 (0.029)	-0.020 (0.019)	1.687* (0.987)
Crony × Low Reg. Qual.	0.213*** (0.049)	0.152*** (0.050)	0.046 (0.034)	0.041 (0.033)

Table 8 (continued)

Fixed-effects model: dependent variable: innovation

	Reg 1	Reg 2	Reg 3	Reg 4
dt_18	−0.256*** (0.055)		−0.084** (0.039)	
dt_19	0.134*** (0.017)	0.146*** (0.050)	0.112*** (0.021)	0.100*** (0.033)
dt_20	0.136*** (0.016)	0.133*** (0.016)	0.103*** (0.018)	0.126*** (0.017)
dt_21	0.109*** (0.015)	0.096*** (0.015)	0.059*** (0.015)	0.078*** (0.014)
N	243	196	243	196
LSDV R-squared	0.986	0.990	0.994	0.996
Within R-squared	0.734	0.761	0.887	0.906
Durbin–Watson	1.185	1.416	1.179	1.499

which are significant provide mostly inconsistent results across the four regressions. The most consistent result is that of the interaction of cronyism with low regulatory quality. It has always a positive effect on innovation, but it is only statistically significant in regressions B1 and B2, and loses significance when we add more control variables in regressions B3 and B4. Another slightly consistent result is that of the interaction of also regulatory quality but this time with PPD. In all the regressions except for B1, it has a negative effect on innovation, but it is statistically significant in regressions B2 and B3 only. This seems to suggest that cronyism could counteract low regulatory quality while PPD can't. Other results that contradict with the main regressions spread across the four regressions without being neither consistent, nor highly significant. Many are significant only at 10% of significance at one or two of the four regressions and insignificant elsewhere, raising more doubts on the reliability of the obtained results. The exception seems to be the interaction of voice and accountability with business association participation which is highly statistically significant in one of the four regressions (B2), but has a negative effect counter to the results obtained in the main regressions.

This shows that the concerns raised earlier on using the fixed effects model were valid. It is to be noted how the business association participation, high official representation in PPD and the PPD proxy variables and their interactions were mostly insignificant. This could be attributed to their time invariant nature that hindered their proper estimation using fixed effects model. However, the results obtained from these four regressions, and despite their contradictory and unreliable results, seem to reinforce the suggestion that cronyism counteracts low regulatory quality and lead (only in this case) to more innovation.

5 Conclusion

The analysis done throughout this paper, with the help of the theoretical discussion and empirical results, suggests that the presence of business associations that are widely-participated help in promoting innovation. They do this also in countries suffering from low institutional settings, and corrects for the shortcomings caused by low regulatory qualities. Businesspeople are more likely to be more aware of the regulatory deficiencies that hinder their innovational efforts. High business participation in these associations guarantees that the policy recommendation input, which they could provide whenever consulted, is very valuable as it reflects the concerns of a big part of businesspeople. PPD play also an important role in correcting institutional deficiencies that could otherwise harm innovation; and this is specifically the case for low government effectiveness. High official representation is a key for such needed outcome. If the government is represented by high officials, it is more likely to be more committed to the process of state–business consultation. In countries suffering from lower government effectiveness, this could be very crucial in guaranteeing that higher state officials would intervene in a way that guarantees that business people's policy recommendations would be implemented by more efficient elements in the bureaucracy or would bypass redtap. This should impact positively on innovation. But, such consultations need an open climate where the government

is accountable to various societal groups and where different interest groups are free to aggregate their power. Otherwise, dialogue and consultation could serve only window-dressing and decorative purposes.

Disturbing as it might be, favoritism in the allocation of resources and cronyism might solve the problem of having institutional deficiencies, especially with regard to low government effectiveness and low regulatory quality. More disturbing is that cronyism seems to be more efficient than PPD in countering low government effectiveness when both highly officially-represented and business-participated PPD and cronyism both exist in a country with low institutional settings. Yet, the observation remains that tackling the reduction of cronyism as an objective, presents PPD as an alternative that could counter low government effectiveness, provided that cronyism is minimized (or in abstract terms, becomes non-existent).

Accounting for the perspective where technological and growth divergence is more possible and past innovation performance affects present levels, gives a slightly different picture. In this one, cronyism still harms innovation and a possible cause for innovational divergence; but it counteracts institutional deficiencies. Voice and accountability is important for unleashing the positive effect of highly-participated business associations on innovation and PPD with high official representation and highly-participated business associations evolve as promoter of innovation regardless of the presence or lack of institutional deficiencies.

Working on improving the institutional settings of a country is an important endeavor and should lead ultimately to creating better conditions that promote innovation. Yet, such efforts should be, expectedly, hard and would need time. This could be expected especially with regard to government effectiveness and regulatory quality. Upgrading the bureaucratic apparatus is hardly an easy task, and so is streamlining regulations. In such conditions, this study suggests that broadly-participated business associations and formal SBR (e.g.: PPD) with high official representation could provide the remedy that would foster innovation, especially if a more open, accountable and democratic system is at operation. Cronyism and crony allocations can't be a long-run mechanism for substituting low institutional qualities, and cronyism by itself is a reflection of institutional deficiencies. It should be remembered that even if cronyism substitutes low government effectiveness and regulatory qualities, its overall effect on innovation is still negative. Thus, to foster innovation both institutional development and the establishment of healthier formal SBR are needed. The latter would be realized with PPD mechanism in which the government is represented by its higher officials and businesspeople are represented by broadly-participated business associations reflecting the interest of a big part of businesspeople.

These findings and arguments fit well the perspective of National Innovation Systems. High business participation in business associations offers an excellent chance for inter-business interactions and communication. Reinforcing the dialogue between state officials and businesspeople intensifies their interaction and fosters their communication. Adding academics and researchers would further the benefits. As the NIS suggests, fostering both inter-firm and public-private interactions should encourage knowledge and information flows, which would on their turn be important for developing, importing, modifying and spreading new technologies. The findings of this paper partially provide empirical evidence in support of NIS, and

especially the role of business associations and public–private collaboration in supporting innovation. Works of that sort are in short supply given the difficulty faced in quantifying NIS, as highlighted in Fagerberg and Srholec (2008).⁶

In addition to the NIS perspective, this paper shows that good governance institutions such as voice and accountability are important for the success of business associations and PPD in upgrading innovational capacities. It also shows that both PPD and business associations could counteract the effect of lower institutional capacities on innovation, especially in the case of lower regulatory quality and government effectiveness.

References

- Achen, C. H. (2000). Why lagged dependent variables can suppress the explanatory power of other independent variables. *Ann Arbor*, 1001, 48106–1248.
- Aghion, P., Harris, C., Howitt, P., & Vickers, J. (2001). Competition, imitation and growth with step-by-step innovation. *The Review of Economic Studies*, 68(3), 467–492.
- Aghion, P., & Howitt, P. (1998). *Endogenous growth theory* (2nd ed.). Cambridge: The MIT Press.
- Alsina, A. (2013). Private equity in the North African Region: Case study of Morocco, Tunisia and Algeria. *European Institute of the Mediterranean Papers* 15.
- Begley, T. M., Khatri, N., & Tsang, E. (2010). Networks and cronyism: A social exchange analysis. *Asia Pacific Journal of Management*, 27, 281–297.
- Bräutigam, D., Rakner, L., & Taylor, S. (2002). Business associations and growth coalitions in Sub-Saharan Africa. *The Journal of Modern African Studies*, 40(04), 519–547.
- Campos, J., & Root, H. (1996). *The key to the Asian miracle: Making shared growth credible*. Washington, DC: Brookings Institution.
- Capello, R., & Lenzi, C. (2014). Spatial heterogeneity in knowledge, innovation, and economic growth nexus: Conceptual reflections and empirical evidence. *Journal of Regional Science*, 54(2), 186–214.
- Cohen, W. M., & Levinthal, D. A. (1989). Innovation and learning: The two faces of R&D. *The Economic Journal*, 99(397), 569–596.
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35, 128–152.
- Crescenzi, R., Gagliardi, L., & Percoco, M. (2013). *The bright side of social capital: How bridging makes Italian provinces more innovative*. *Geography, Institutions and Regional Economic Performance* (pp. 143–164). Berlin: Springer.
- Desai, R. M. (2007). *How cronyism harms the investment climate*. Washington DC: The Brookings Institution.
- Diwan, I., Keefer, P., & Schiffbauer, M. (2014). *On top of the Pyramids: Cronyism and private sector growth in Egypt*, Working Paper. Washington DC: World Bank.
- El-Elj, M. (2012). Innovation in Tunisia: Empirical analysis for industrial sector. *Journal of Innovation Economics and Management*, 1(9), 183–197.
- Enderwick, P. (2005). What's bad about crony capitalism? *Asian Business and Management*, 4(2), 117–132.
- Fagerberg, J., & Srholec, M. (2008). National innovation systems, capabilities and economic development. *Research Policy*, 37, 1417–1435.

⁶ Fagerberg and Srholec (2008) suggested a proxy for NIS depending on the principal component factors method. It incorporated “patenting, scientific publications, ICT infrastructure, ISO 9000 certifications and access to finance” besides “education” and then tested the effect of this proxy on economic growth finding a positive effect. This proxy, unfortunately, couldn't be used in testing the effect of NIS on innovation because it measures aspects which are often used in measuring innovation itself. In fact, it overlaps with components of the innovation proxy used in this paper. It also neglects important aspects of NIS of much relevance to the research conducted here such as the role of SBR.

- Haggard, S., Maxfield, S., & Schneider, B. R. (1997). Theories of business and business-state relations. In S. Maxfield & B. R. Schneider (Eds.), *Business and the state in developing countries*. Ithaca: Cornell University Press.
- Hausmann, R., & Rodrik, D. (2003). Economic development as self discovery. *Journal of Development Economics*, 72(2), 603–633.
- Hausmann, R., Rodrik, D., & Sabel, C. (2008). *Reconfiguring industrial policy: A framework with an application to South Africa*. Cambridge: Harvard Kennedy School of Government: Faculty Research Working Papers Series 08-031.
- Herzberg, B. & Wright, A. (2005). Competitiveness partnerships—Building and maintaining Public–Private Dialogue to improve the investment climate: A resource drawn from the review of 40 countries' experience. The World Bank Policy Research, Working Paper 3683.
- Herzberg, B., & Wright, A. (2006). *The public private dialogue handbook: A toolkit for business environment reformers*. USA: DFID, the WB Group.
- Kang, D. C. (2003). Transaction costs and Crony Capitalism in East Asia. *Comparative Politics*, 35(4), 439–458.
- Kaufmann, D., Kraay, A. & Mastruzzi, M. (2009). Governance Matters VIII: Aggregate and Individual Governance Indicators 1996–2008. The World Bank Policy Research, Working Paper 4978.
- Keefer, P. (2007). Governance and economic growth. In L. A. Winters & S. Yusuf (Eds.), *Dancing with giants: China, India and the global economy*. Washington: The World Bank and Institute of Policy Studies.
- Keele, L., & Kelly, N. J. (2005). Dynamic models for dynamic theories: The ins and outs of lagged dependent variables. *Political Analysis*, 14(2), 186–205.
- Leftwich, A. & te Velde, D. (2010). State–business relations and economic growth in Sub-Saharan Africa: A review of case studies in Ghana, Mauritius, South Africa and Zambia, IPPG Discussion Paper Series 48.
- Mazumdar, S. (2008). Crony capitalism: Caricature or category? Institute for studies in Industrial Development, Working Paper 02.
- Moon, C., & Prasad, R. (1994). Beyond the developmental state: Networks, politics, and institutions. *Governance: An International Journal of Policy and Administration*, 7(4), 360–386.
- Mueller, D. (2003). *Public choice III*. Cambridge: Cambridge University Press.
- OECD. (1997). *National innovation systems*. Paris: OECD Publishing.
- OECD. (2015). *New approaches to SME and entrepreneurship financing: Broadening the range of instruments*. Paris: OECD Publishing.
- Olson, M. (1982). *The rise and decline of nations: Economic growth stagflation and social rigidities*. New Haven: Yale University Press.
- Putnam, R. D., Leonardi, R., & Nanetti, R. Y. (1993). *Making democracy work: Civic institutions in modern Italy*. NJ: Princeton University Press.
- Romijn, H. A., & Caniëls, M. C. (2011). Pathways of technological change in developing countries: Review and new agenda. *Development Policy Review*, 29(3), 359–380.
- Rothstein, B., & Stolle, D. (2008). The state and social capital: An institutional theory of generalized trust. *Comparative Politics*, 40, 441–467.
- Sabry, M. I. (2013). *State business relations: Networks, institutions and their effect on growth and cronyism*. Munich: Verlag Dr Hut.
- Sabry, M. I. (2017a). Informal state–business connections, institutions and economic growth. *Economia Politica*, 34(2), 233–258.
- Sabry, M. I. (2017b). Taming cronyism!: The role of public private dialogues. *International Journal of Social Economics*, 44(12), 1622–1638.
- Sabry, M. I. (2018). Cronyism as an outcome of institutional settings: The case of pre-2011 Egypt. *Topics in Middle Eastern and African Economies*, 20(1), 85–111.
- Sapir, A. (1977). Use of the Durbin Watson Statistic with lagged dependent variables. *Metroeconomica*, 29(1–3), 169–172.
- Schneider, B. R. (2010). *Business–Government interaction in Policy Councils in Latin America: Cheap talk, expensive exchanges, or collaborative learning?* Inter-American Development Bank Working Paper Series, No. IDB-WP-167.
- Schneider, B. R., & Maxfield, S. (1997). Business, the state, and economic performance in developing countries. *Business and the state in developing countries* (pp. 3–25). New York: Cornell University Press.

- Schwab, K., & Sala-i-Martin, X. (2016). *The Global Competitiveness Report 2016–2017*. Switzerland: World Economic Forum.
- te Velde, D. (2009). Analyzing the economics of State–Business Relations: A summary guide, IPPG Discussion Paper Series 23B.
- Ulke, H. (2004). R&D, innovation, and economic growth: An empirical analysis. IMF Working Paper WP/04/185.
- United Nations Industrial Development Organization (UNIDO). (2000). *Public–Private partnerships for economic development and competitiveness: With special reference to the African experience*. Vienna: UNIDO.
- Watkins, A., Papaioannou, T., Mugwagwa, J., & Kale, D. (2015). National innovation systems and the intermediary role of industry associations in building institutional capacities for innovation in developing countries: A critical review of the literature. *Research Policy*, 44, 1407–1418.

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